Практическое занятие 12

Формула Тейлора

Представление основных элементарных функций по формуле Тейлора (Маклорена)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \bar{o}(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} + \bar{o}(x^{2n+2})$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \cdot \frac{x^{2n}}{(2n)!} + \bar{o}(x^{2n+1})$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n+1} \cdot \frac{x^n}{n} + \bar{o}(x^n)$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + \bar{o}(x^n)$$

$$(1+x)^{-1} = \frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n \cdot x^n + \bar{o}(x^n)$$

$$(1-x)^{-1} = \frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \bar{o}(x^n)$$

$$\arcsin x = x + \frac{x^3}{6} + \bar{o}(x^4)$$

$$\operatorname{arctg} x = x - \frac{x^3}{3} + \bar{o}(x^4)$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + \bar{o}(x^4)$$

Примеры

Представить функцию f(x) по формуле Тейлора в окрестности точки a.

1) $f(x) = \cos 3x\,,\ \ a = 0.$

$\cos 3x = 1 - \frac{(3x)^2}{2!} + \frac{(3x)^4}{4!} - \cdots + (-1)^n \frac{(3x)^{2n}}{(2n)!} + \bar{o}(x^{2n+1}) = 1 - \frac{3^2}{2!}x^2 + \frac{3^4}{4!}x^4 - \cdots + (-1)^n \frac{3^{2n}}{(2n)!}x^{2n} + \bar{o}(x^{2n+1})$

2) $f(x) = \ln x,\ \ a = 2.$

$\ln x = \ln\big(2 + (x-2)\big) = \ln\left(2\left(1 + \frac{x-2}{2}\right)\right) = \ln 2 + \ln\left(1 + \frac{x-2}{2}\right) = \ln 2 + \frac{x-2}{2} - \frac{1}{2}\left(\frac{x-2}{2}\right)^2 + \frac{1}{3}\left(\frac{x-2}{2}\right)^3 - \cdots + \frac{(-1)^{n+1}}{n}\left(\frac{x-2}{2}\right)^n + \bar{o}((x-2)^n) = \ln 2 + \frac{1}{2}(x-2) - \frac{1}{2*2^2}(x-2)^2 + \frac{1}{3*2^3}(x-2)^3 - \cdots + \frac{(-1)^{n+1}}{n\,2^n}(x-2)^n + \bar{o}((x-2)^n)$

3) $f(x) = \frac{1}{2x-3},\ \ a = 2.$

$\frac{1}{2x-3} = \frac{1}{2(x-2)+1} = \frac{1}{1+2(x-2)} = 1 - \big(2(x-2)\big) + \big(2(x-2)\big)^2 - \cdots + (-1)^n\big(2(x-2)\big)^n + \bar{o}((x-2)^n) = 1 - 2(x-2) + 2^2(x-2)^2 - \cdots + (-1)^n * 2^n * (x-2)^n + \bar{o}((x-2)^n)$

4) $f(x) = \frac{1}{x^2-3x+2},\ \ a = 0$

$\frac{1}{x^2-3x+2} = \frac{1}{(x-2)(x-1)} = \frac{1}{x-2} - \frac{1}{x-1} = -\frac{1}{2}\frac{1}{1-\frac{x}{2}} + \frac{1}{1-x} = \big(1 + x + x^2 + \cdots + x^n + \bar{o}(x^n)\big) - \frac{1}{2}\left(1 + \frac{x}{2} + \left(\frac{x}{2}\right)^2 + \cdots + \left(\frac{x}{2}\right)^n + \bar{o}(x^n)\right) = \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2^2}\right)x + \left(1 - \frac{1}{2^3}\right)x^2 + \cdots + \left(1 - \frac{1}{2^{n+1}}\right)x^n + \bar{o}(x^n)$

5) $f(x) = \frac{2x^2-7x-1}{2x+1},\ \ a = -1,\ \ f^{(n)}(-1) = ?$

$\frac{2x^2-7x-1}{2x+1} = x - 4 + \frac{3}{2x+1} = (x+1) - 5 + \frac{3}{2(x+1)-1} = -5 + (x+1) - \frac{3}{1-2(x+1)} = -5 + (x+1) - 3\left(1 + \big(2(x+1)\big) + \big(2(x+1)\big)^2 + \cdots + \big(2(x+1)\big)^n\right) + \bar{o}((x+1)^n) = -8 - 5(x+1) - 3*2^2(x+1)^2 - \cdots - 3*2^n(x+1)^n + \bar{o}((x+1)^n).$

$$\frac{f^{(n)}(-1)}{n!} = -3*2^n, \qquad f^{(n)}(-1) = -3*2^n\,n!$$

Вычисление пределов.

1) $\lim\limits_{x\to0}\frac{\cos 2x-e^{-2x^2}}{x^4}=\left(\frac{0}{0}\right)=\lim\limits_{x\to0}\frac{-\frac{4}{3}x^4+\bar{o}(x^4)}{x^4}=\lim\limits_{x\to0}\left(-\frac{4}{3}+\bar{o}(1)\right)=-\frac{4}{3}$

$$\cos 2x=1-\frac{(2x)^2}{2!}+\frac{(2x)^4}{4!}+\bar{o}(x^4)=1-2x^2+\frac{2}{3}x^4+\bar{o}(x^4)$$

$$e^{-2x^2}=1+(-2x^2)+\frac{(-2x^2)^2}{2!}+\bar{o}(x^4)=1-2x^2+2x^4+\bar{o}(x^4)$$

Числитель $=1-2x^2+\frac{2}{3}x^4-1+2x^2-2x^4+\bar{o}(x^4)=-\frac{4}{3}x^4+\bar{o}(x^4)$

2) $\lim\limits_{x\to0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}=(1^\infty)=\lim\limits_{x\to0}e^{\frac{1}{x^2}\ln\left(\frac{\sin x}{x}\right)}=e^{-\frac{1}{6}}=\frac{1}{\sqrt[6]{e}}$

$$\lim_{x\to0}\frac{\ln\left(\frac{\sin x}{x}\right)}{x^2}=\left(\frac{0}{0}\right)=\lim_{x\to0}\frac{\ln\left(\frac{1}{x}\left(x-\frac{x^3}{3!}+\bar{o}(x^4)\right)\right)}{x^2}$$

$$=\lim_{x\to0}\frac{\ln\left(1-\frac{x^2}{6}+\bar{o}(x^3)\right)}{x^2}=\lim_{x\to0}\frac{-\frac{x^2}{6}+\bar{o}(x^3)}{x^2}=-\frac{1}{6}$$

3) $\lim\limits_{x\to0}\left(\frac{1}{x}-\frac{1}{\sin x}\right)=(\infty-\infty)=\lim\limits_{x\to0}\frac{\sin x-x}{x\sin x}=\lim\limits_{x\to0}\frac{x-\frac{x^3}{3!}+\bar{o}(x^3)-x}{x^2}=$
$\lim\limits_{x\to0}\frac{-\frac{x^3}{6}+\bar{o}(x^3)}{x^2}=\lim\limits_{x\to0}\left(-\frac{x}{6}+\bar{o}(x)\right)=0$

4) $\lim\limits_{x\to0}\left(\frac{1}{x^2}-\frac{1}{\sin^2 x}\right)=(\infty-\infty)=\lim\limits_{x\to0}\frac{\sin^2 x-x^2}{x^2\sin^2 x}=$
$\lim\limits_{x\to0}\frac{(\sin x-x)(\sin x+x)}{x^4}=\lim\limits_{x\to0}\frac{(x-\frac{x^3}{3!}+\bar{o}(x^3)-x)(x+\bar{o}(x)+x)}{x^4}=$
$\lim\limits_{x\to0}\frac{\left(-\frac{x^3}{6}+\bar{o}(x^3)\right)(2x+\bar{o}(x))}{x^4}=\lim\limits_{x\to0}\frac{-\frac{x^4}{3}+\bar{o}(x^4)}{x^4}=-\frac{1}{3}$

5) $\lim\limits_{x\to0}\frac{1-(\cos x)^{\sin x}}{x^3}=\left(\frac{0}{0}\right)=\lim\limits_{x\to0}\frac{1-1+\frac{x^3}{2}+\bar{o}(x^3)}{x^3}\lim\limits_{x\to0}\frac{\frac{x^3}{2}+\bar{o}(x^3)}{x^3}=\frac{1}{2}$

$$(\cos x)^{\sin x} = e^{\sin x \ln \cos x} = e^{(x+\bar{o}(x))\ln\left(1-\frac{x^2}{2!}+\bar{o}(x^2)\right)}$$
$$= e^{(x+\bar{o}(x))\left(-\frac{x^2}{2}+\bar{o}(x^2)\right)} = e^{-\frac{x^3}{2}+\bar{o}(x^3)} = 1-\frac{x^3}{2}+\bar{o}(x^3)$$

6) $\lim\limits_{x\to 0}\frac{\sin 3x+\sqrt{1+2x}-e^{4x}}{\cos 3x-\sqrt{1+2x}+\ln(1+x)} = \left(\frac{0}{0}\right) = \lim\limits_{x\to 0}\frac{-\frac{17}{2}x^2+\bar{o}(x^2)}{-\frac{9}{2}x^2+\bar{o}(x^2)} =$

$\lim\limits_{x\to 0}\frac{-\frac{17}{2}+\bar{o}(1)}{-\frac{9}{2}+\bar{o}(1)} = \frac{17}{9}$

$$\sin 3x = 3x + \bar{o}(x^2)$$

$$\sqrt{1+2x} = (1+2x)^{\frac{1}{2}} = 1+\frac{1}{2}(2x)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(2x)^2+\bar{o}(x^2)$$
$$= 1+x-\frac{1}{2}x^2+\bar{o}(x^2)$$

$$e^{4x} = 1+(4x)+\frac{(4x)^2}{2!}+\bar{o}(x^2) = 1+4x+8x^2+\bar{o}(x^2)$$

$$\cos 3x = 1-\frac{(3x)^2}{2!}+\bar{o}(x^2) = 1-\frac{9}{2}x^2+\bar{o}(x^2)$$

$$\ln(1+x) = x-\frac{x^2}{2}+\bar{o}(x^2)$$

Числитель = $3x+1+x-\frac{1}{2}x^2-1-4x-8x^2+\bar{o}(x^2) = -\frac{17}{2}x^2+\bar{o}(x^2)$

Знаменатель = $1-\frac{9}{2}x^2-1-x+\frac{1}{2}x^2+x-\frac{x^2}{2}+\bar{o}(x^2) = -\frac{9}{2}x^2+\bar{o}(x^2)$